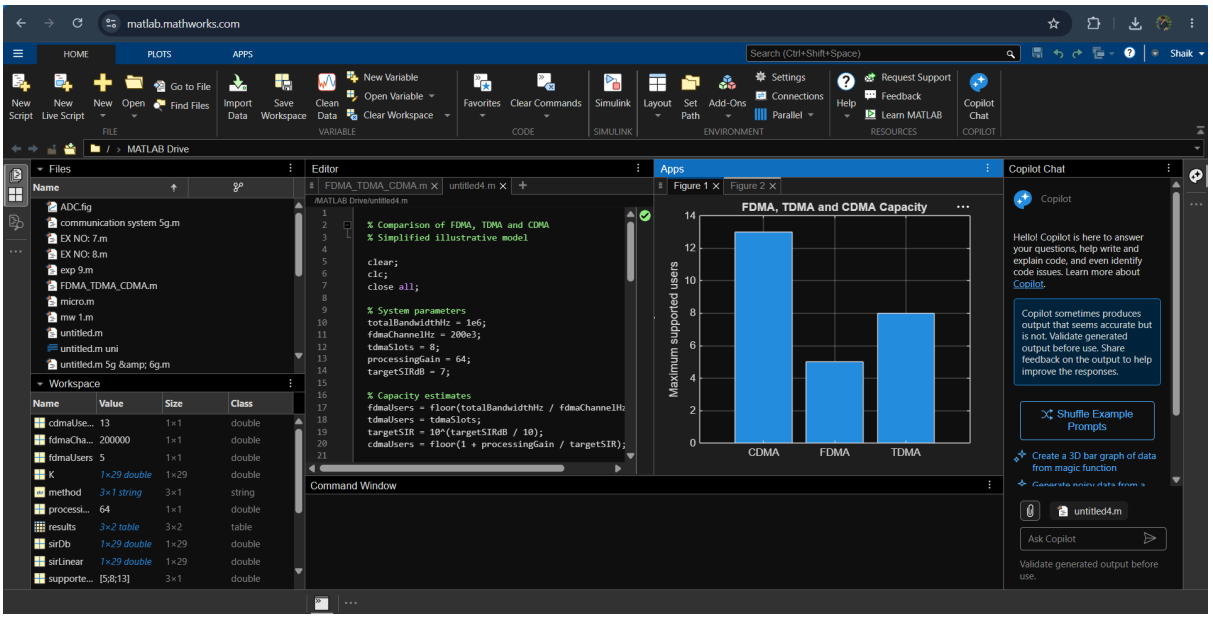
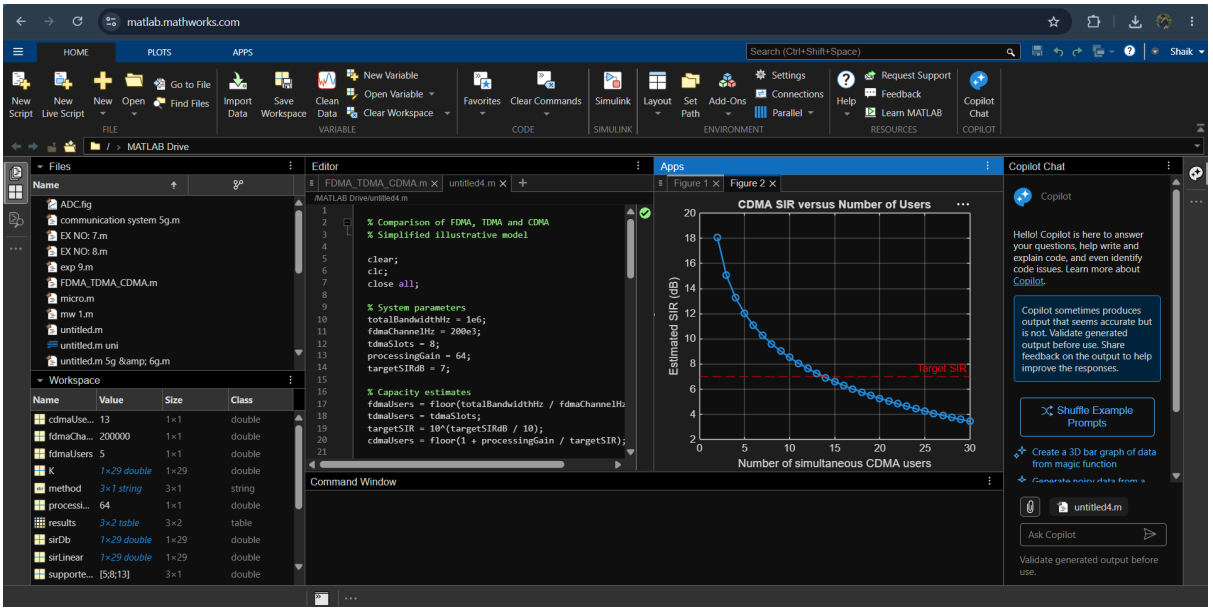
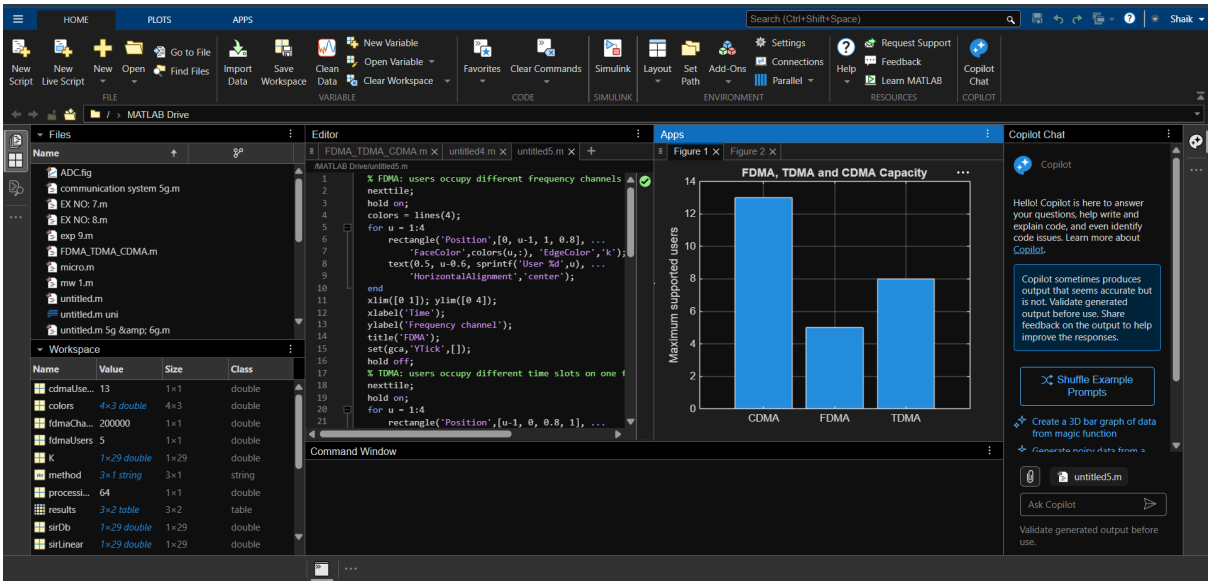
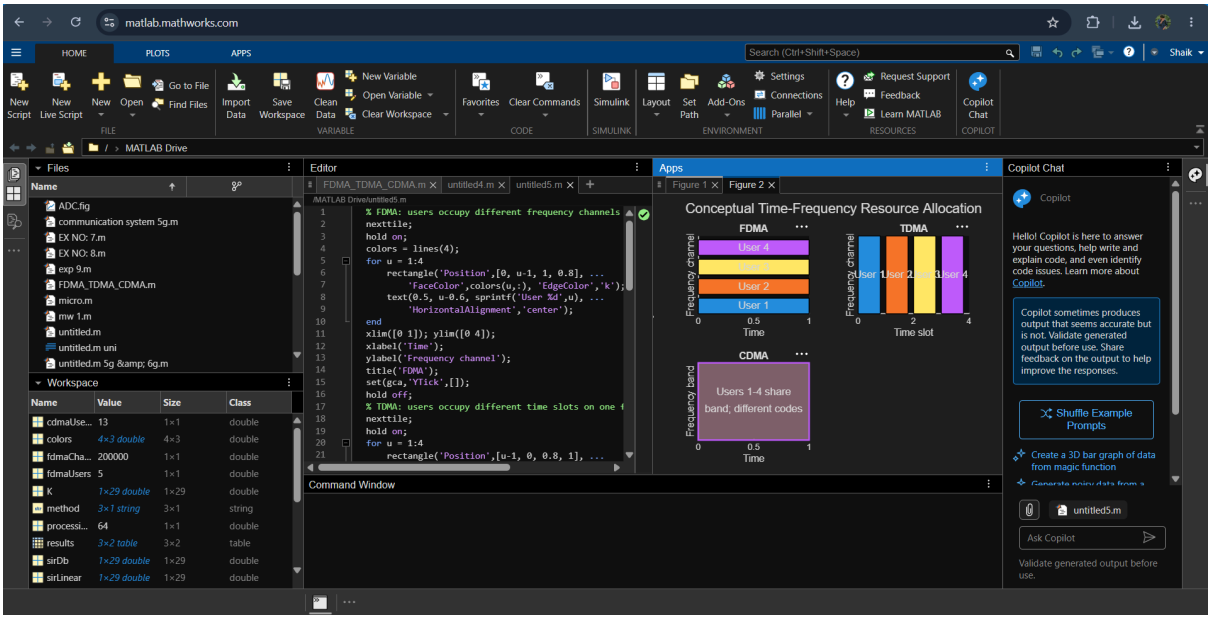
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FDMA_TDMA_CDMA.m
untitled4.m
% Comparison of FDMA, TDMA and CDMA
% Simplified illustrative model
clear;
clc;
close all;
% System parameters
totalBandwidthHz = 1e6;
fdmaChannelHz = 200e3;
tdmaSlots = 8;
processingGain = 64;
targetSIRdB = 7;
% Capacity estimates
fdmaUsers = floor(totalBandwidthHz / fdmaChannelHz);
tdmaUsers = tdmaSlots;
targetSIR = 10^(targetSIRdB / 10);
cdmaUsers = floor(1 + processingGain / targetSIR);
FDMA, TDMA and CDMA Capacity
Maximum supported users
CDMA
FDMA
TDMA
Command Window
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CDMA SIR versus Number of Users
Estimated SIR (dB)
Target SIR
Number of simultaneous CDMA users
Command Window
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micro.m
mw 1.m
untitled.m
untitled.m uni
untitled.m 5g & 6g.m
Workspace
Name
Value
Size
Class
cdmaUse... 13 1×1 double
colors 4×3 double 4×3 double
fdmaCha... 200000 1×1 double
fdmaUsers 5 1×1 double
K 1×29 double 1×29 double
method 3×1 string 3×1 string
processi... 64 1×1 double
results 3×2 table 3×2 table
sirDb 1×29 double 1×29 double
sirLinear 1×29 double 1×29 double
Editor
FDMA_TDMA_CDMA.m
untitled4.m
untitled5.m
% FDMA: users occupy different frequency channels
nexttile;
hold on;
colors = lines(4);
for u = 1:4
rectangle('Position',[0, u-1, 1, 0.8], ...
'FaceColor',colors(u,:), 'EdgeColor','k');
text(0.5, u-0.6, sprintf('User %d',u), ...
'HorizontalAlignment','center');
end
xlim([0 1]); ylim([0 4]);
xlabel('Time');
ylabel('Frequency channel');
title('FDMA');
set(gca,'YTick',[]);
hold off;
% TDMA: users occupy different time slots on one f
nexttile;
hold on;
for u = 1:4
rectangle('Position',[u-1, 0, 0.8, 1], ...
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Figure 1
Figure 2
FDMA, TDMA and CDMA Capacity
Maximum supported users
CDMA
FDMA
TDMA
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Figure 1
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Conceptual Time-Frequency Resource Allocation
FDMA
TDMA
CDMA
User 4
User 3
User 2
User 1
Frequency channel
Time
Time slot
Frequency band
Users 1-4 share band; different codes
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